

TECHNICAL REPORT

VECTRA-X

Uncertainty-Aware Triage Intelligence for Vector-Borne Disease Response in Resource-Limited Humanitarian Settings

FIT Competition 2026 | Track IV: AI-based Vector-Borne Disease Prediction
Evidence generated from a self-contained notebook and the official CSV

Executive Summary

VECTRA-X reframes vector-borne disease prediction as an operational triage problem. It preserves co-infection, separates feature timing, exposes target leakage, communicates uncertainty, and creates transparent review and resource queues.

Primary finding

Extra Trees ranked first by training-only OOF macro PR-AUC. Calibrated PRE_LAB held-out macro F1 was 0.535, micro F1 0.851, and macro PR-AUC 0.582.

Safety limitation

Yellow fever recall was 0/3 and typhoid recall was 1/7. The system is decision-support, not diagnosis, and requires prospective validation.

1. Humanitarian Problem and Intended Use

Outbreak teams must decide who needs urgent review, who should receive scarce confirmatory tests, and when model uncertainty is too high for automated priority. VECTRA-X supports these decisions without replacing clinician assessment or confirmatory testing.

2. Related Work and Research Gap

Most febrile-illness models optimize one disease or mutually exclusive classes. VECTRA-X addresses stage-aware, leakage-audited, multi-label triage with uncertainty and scarce-resource prioritization.

3. Data and Problem Formulation

Property	Observed value	Consequence
Patients	300	Use uncertainty intervals
Variables	109	Audit schema and timing
Health centers	2	Stress-test facility transfer
Multi-label patients	158	Use multi-label learning
Active labels	5	Report macro and per-label metrics

4. Leakage-Safe Methodology

The untouched multi-label holdout is created immediately after target extraction. All learned schema, imputation, model selection, thresholds, calibration, and conformal thresholds use training data only.

- PRE_LAB: demographics, symptoms, history, and available vital signs.
- LAB_AWARE: adds ordered laboratory and rapid-test information.
- FULL: includes target-restatement risks and is a research-only leakage demonstration.

5. Model Comparison

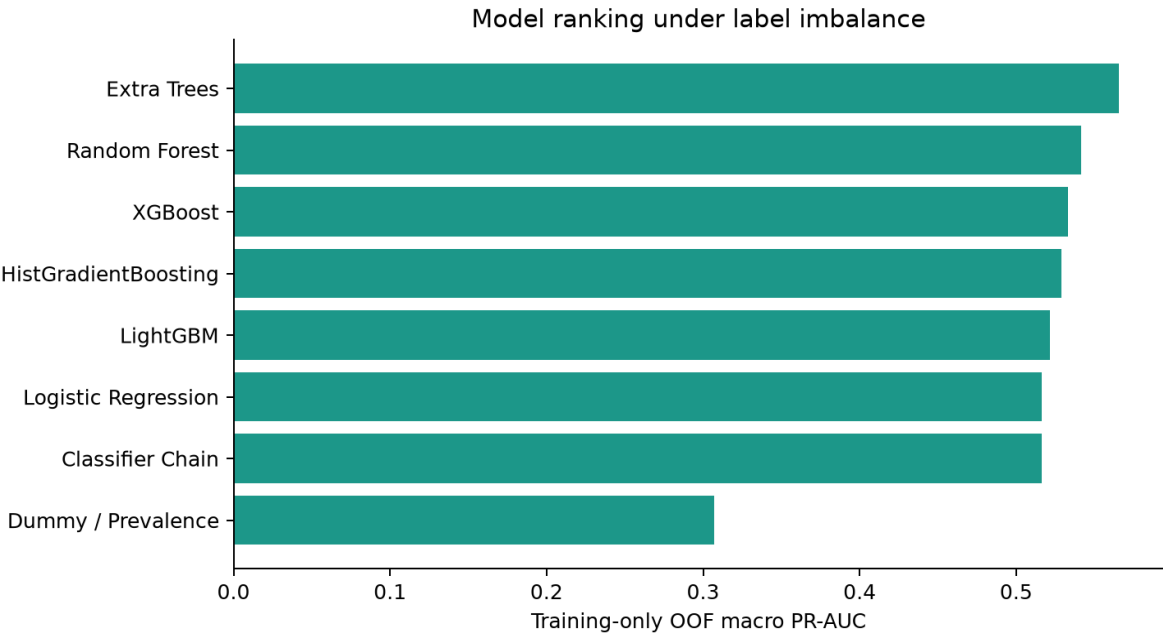


Figure 1. Training-only OOF model ranking by macro PR-AUC.

Model	OOF PR-AUC	Runtime
Extra Trees	0.566	22.4s
Random Forest	0.541	21.6s
XGBoost	0.533	5.2s
HistGradientBoosting	0.529	14.6s
LightGBM	0.521	4.7s
Logistic Regression	0.516	0.7s
Classifier Chain	0.516	0.6s
Dummy / Prevalence	0.307	0.5s

Macro PR-AUC ranks candidates under imbalance. Final triage suitability also considers recall, false negatives, calibration, stability, and workload.

6. Held-Out Results

Track	Macro F1	Micro F1	Macro recall	Macro PR-AUC
PRE_LAB	0.591	0.782	0.792	0.589
LAB_AWARE	0.613	0.815	0.740	0.595
FULL	0.662	0.843	0.754	0.682

FULL performance is higher but operationally invalid because it includes post-diagnosis signal. PRE_LAB remains the honest early-triage track.

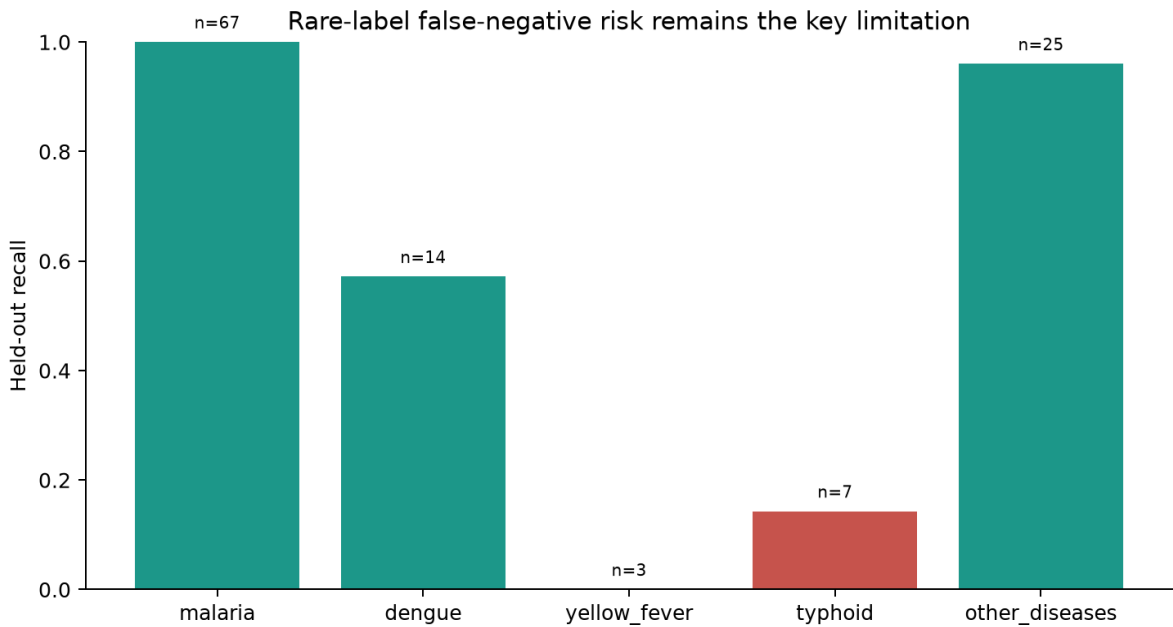


Figure 2. Held-out recall with positive support counts.

Label	Support	Precision	Recall	F1	FNR
malaria	67	0.870	1.000	0.931	0.000
dengue	14	0.471	0.571	0.516	0.429
yellow_fever	3	0.000	0.000	0.000	1.000
typhoid	7	1.000	0.143	0.250	0.857
other_diseases	25	1.000	0.960	0.980	0.040

7. Metrics for Real-World Triage

Question	Metric	Operational purpose
Rank rare diseases?	Macro PR-AUC	Limits common-label dominance
Which labels are missed?	Recall and FNR	Exposes false negatives
How much review work?	Precision and NNR	Quantifies staff burden
Can probabilities guide priority?	Brier and ECE	Tests probability trust
Does testing policy help?	Decision-curve net benefit	Compares model and default strategies
Will evidence transfer?	CI, repeated CV, center transfer	Exposes instability

8. Calibration, Uncertainty, and Caution Sets

Label	Support	Brier raw	Brier calibrated	ECE raw	ECE calibrated
malaria	67	0.109	0.114	0.064	0.075
dengue	14	0.130	0.126	0.087	0.081
yellow_fever	3	0.036	0.037	0.049	0.000
typhoid	7	0.104	0.078	0.105	0.049
other_diseases	25	0.049	0.015	0.161	0.010

Calibration is label-dependent. Entropy and probability margin route uncertain cases to human review. Label-wise conformal sets are a caution mechanism, not a joint multi-label guarantee.

Label	Test positives	Coverage	Inclusions
malaria	67	1.000	77
dengue	14	0.929	36
yellow_fever	3	1.000	77
typhoid	7	0.286	11
other_diseases	25	0.960	24

9. Fairness and Center Robustness

Subgroup results are warning diagnostics, not proof of fairness. Facility transfer is a stronger stress test and shows that prospective deployment requires center-specific validation and recalibration.

Test Center	N Test	Macro F1	Recall Dengue	Recall Yellow Fever	Recall Typhoid
CMA de DO	147	0.350	0.000	0.000	0.000
CMA de DAFRA	153	0.368	0.000	0.000	0.000

10. Triage and Resource Workflow

Each case receives separate prediction, uncertainty, triage priority, and recommended human action fields. Resource simulation makes unserved demand visible.

Resource	Demand	Capacity	Unserved
rapid_tests	33	50	0
beds	7	12	0

Resource	Demand	Capacity	Unservd
staff_review_slots	49	80	0

Policy	Flags	True-positive flags	Number needed to review	Flags/patient
performance	119	100	1.19	1.55
safety	122	100	1.22	1.58
operational	106	93	1.14	1.38

11. Ethics, Privacy, and Limitations

- No treatment recommendation is produced, and clinician assessment remains authoritative.
- Rare-label estimates are unstable, especially yellow fever and typhoid.
- Public exports use case IDs and exclude UUIDs and truth labels.
- Fairness estimates are warning diagnostics, not evidence of equity.
- Triage weights and resource policies require clinician co-design.

12. Conclusion

The strongest contribution is the workflow around an honest pre-lab model: multi-label framing, leakage control, uncertainty-aware human review, and transparent resource prioritization. Future work requires prospective temporal and multi-center validation, larger rare-label cohorts, facility recalibration, clinician usability testing, and outcome-based resource optimization.

Appendix A. Confidence Intervals

Metric	Estimate	95% CI low	95% CI high	Draws
macro_f1	0.535	0.446	0.626	500
micro_f1	0.851	0.794	0.904	500
macro_recall	0.535	0.460	0.622	500
macro_pr_auc	0.582	0.518	0.728	500

Appendix B. Reproducibility

Python 3.12.6; random state 42; CSV SHA-256
365344fe8ce553288f952c6d32771cdde2b7c633006538ddb1d6c82de325efb1. The notebook was executed in a clean folder containing only the notebook and CSV.

References

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